BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Comparing Algorithms and Consolidation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How does merge sort combine two sorted lists?

2. Which storage technology does a DVD use?
   A) Optical
   B) Magnetic
   C) Solid-state
   D) Volatile RAM

3. "FROM is the clause that specifies the required output columns." Correct this. Explain why.

4. Code has a missing bracket. "A syntax error is a wrong result from code that follows the language's rules." Correct this. What needs fixing?

Today's objectives

Explain the steps and required conditions for comparing algorithms and consolidation. Apply the algorithm or method to supplied data.

Key vocabulary - Read, cover, recall

Searching
Locating a target in a data set.

Sorting
Arranging items into a specified order.

Prerequisite
Condition required for a method to work correctly.

Divide and conquer
Solving a problem by splitting into smaller parts.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Search versus sort

Searching locates a target; sorting rearranges values into a specified order.

Choose a search

Linear works on unsorted data; binary needs sorted data but often makes fewer comparisons for large lists.

Compare sorting approaches

Bubble swaps adjacent out-of-order pairs; insertion places the next key into a sorted prefix; merge divides and combines ordered sublists.

Explain from evidence

For examination questions, name the method, trace the data and justify any prerequisite or limitation.

I Do - Worked model

A school stores an unsorted list of pupil IDs. Choose a suitable search, then explain what would change if the list were sorted. [5 marks]

- Identify the starting condition
- The original list is unsorted.
- Linear search works without sorting and checks entries in sequence.
- If sorted, binary search could discard half of the active range after each comparison and generally need fewer checks for large lists. [5]

We Do - Guided practice

1. A sorted array contains 128 items. Which search is appropriate and why? [4]

2. A student says bubble and insertion sort are identical. Compare the two. [4]

3. Explain the difference between searching for a value and sorting data. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict which search algorithm may be used for a sorted list of common network protocols.

```
array protocols = ["DNS","HTTP","HTTPS","SMTP"]
sorted = true
if sorted then
print("Binary search possible")
else
print("Use linear search")
endif
```

You Do - Independent exam questions

Q2. Choose a search algorithm for an unsorted list and explain the choice. [3]

Q3. State the characteristic action of each sorting algorithm: bubble, insertion and merge. [3]

Q4. Explain why binary search may outperform linear search on a large sorted array. [4]

Exit Ticket - True or False?

Binary search can safely discard half only when data is sorted.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Compare the next item in each list. Take the smaller. Repeat, then add any items left over.

2. A - Optical

3. The table to get data from.

4. A syntax error breaks the language's rules and stops the affected code being translated or run. Fix it, then trace and test the logic.

I Do - Explain the model

Identify the starting condition

The original list is unsorted.

Linear search works without sorting and checks entries in sequence.

If sorted, binary search could discard half of the active range after each comparison and generally need fewer checks for large lists. [5]

We Do - Model answers

1. Binary search [1] as sorting prerequisite is met [1]; midpoint comparisons halve the range [1], generally using fewer comparisons than linear search for large data [1].

2. Bubble compares/swaps neighbouring elements [1] over repeated passes [1]; insertion builds a sorted prefix [1] by shifting and inserting a key [1].

3. Search aims to locate a value/index [1]; sort reorders all items by a key [1]; binary search may require sorting first [1].

Code First - Answer

The sorted flag is true, so Binary search possible. If false, Use linear search. DNS resolves domain names to IP addresses; HTTPS secures web traffic using TLS encryption. Binary search depends on sorted input so it can safely eliminate half the remaining range. This code chooses an approach; it does not implement the search.

You Do - Self-mark

Q2: Linear search [1]; compares items sequentially [1] without needing the list sorted [1].

Q3: Bubble: adjacent swaps [1]; insertion: insert key into sorted prefix [1]; merge: split then merge ordered sublists [1].

Q4: Binary chooses midpoint [1] and discards half each unsuccessful check [1]; linear may inspect elements individually [1], so binary generally uses fewer checks for large sorted arrays [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.